Activity Recognition Using a Multi-head Convolution Neural Network Integrated with Attention Mechanism

Yousef Hossni and Murad Khan

Abstract Human Activity Recognition (HAR) has emerged as a critical research domain with applications spanning healthcare monitoring to smart environments. Despite significant advances in deep learning approaches for HAR, fundamental challenges persist in modeling the inherent variability of human movements and the sequential nature of physical activities. In this paper, we present a Multi-Head Convolutional Neural Network with an attention mechanism (MH-CNN) that addresses these limitations by dynamically focusing on relevant sensor inputs. Our approach integrates four parallel convolutional pathways with attention mechanisms, enabling the model to process different feature subspaces while maintaining computational efficiency simultaneously. Experiments conducted on the challenging Opportunity dataset (Roggen in UCI Mach Learn Repos, 2010. https://doi.org/10.24432/C5M027) [6] reveal that our architecture achieves superior performance (99.78% test accuracy and F1-score), outperforming conventional CNN approaches (99.75%) and single-head attention models (99.73%). While the accuracy improvement of 0.03% may appear modest, it represents approximately 30 additional correctly classified instances in our test set of approximately 10,000 samples, which is meaningful in safety–critical HAR applications where misclassifications can have significant consequences. Interestingly, we identify and analyze an unexpected phenomenon where single- head attention slightly underperforms compared to standard CNNs, while multi- head configurations provide substantial improvements. These findings demonstrate that our architecture effectively captures the complex temporal dependencies in human activities, offering a promising direction for next- generation HAR systems in real-world applications.

Keywords Activity recognition · Deep learning · CNN · Multi-head · Attention

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1 Introduction

Wearable sensors have transformed human activity recognition (HAR) into a vital research domain with applications ranging from healthcare monitoring to smart home environments. Despite significant progress through deep learning approaches [8, 11], HAR implementations continue to encounter fundamental obstacles, including variability in human movement patterns, the sequential nature of physical activities, and the disproportionate contributions of various sensor inputs to recognition accuracy [5, 10]. To rigorously evaluate models against these challenges, benchmark datasets like the Opportunity dataset [6] are often utilized. This widely adopted benchmark is known for its complexity, featuring multi-modal sensor data (primarily body-worn inertial sensors) capturing 18 distinct, often fine-grained daily living activities performed by multiple subjects, along with frequent transitional or undefined (NULL) states, making it particularly demanding for recognition algorithms.

Conventional HAR methodologies require substantial domain expertise for feature engineering and inadequately model the sequential nature of human activities. Early deep learning architectures offered improvements—CNNs extracted spatial patterns from sensor windows, while RNNs showed potential for temporal modeling—yet still struggled with distinguishing critical components within movement sequences.

Attention mechanisms have revolutionized HAR by enabling models to focus dynamically on relevant input data. Zhang et al. [11] introduced one of the first multi-head attention approaches in 2019, demonstrating how parallel processing of sensor data could significantly improve recognition accuracy. Wang et al. [8] confirmed that deep learning approaches enhance performance by automatically extracting hierarchical features from raw sensor data. Subsequent innovations included Yin et al. [10] ConvBLSTM-PMwA model (2022), which fused 1-D CNNs and bi-directional LSTMs with attention mechanisms to handle noisy smartphone sensor data.

Recent architectural advances include Mekruksavanich and Jitpattanakul [5] hybrid CNN with channel attention mechanisms (2023), Thakur et al. [7] six-head CNN, and Wei and Wang [9] TCN-Attention-HAR employing dilated causal convolutions for capturing long-range temporal dependencies. Additionally, Zhang et al. [12] utilized squeeze and excitation modules as attention mechanisms in DeepConvLSTM, while Ge et al. [3] combined CNNs with Vision Transformers in their ConTransEn model. Other notable contributions include Chandramouli et al. [2] CNN-BiLSTM hybrid, Abdellatef et al. [1] HARCNN, and Khan and Hossni [4] comparative analysis of attention-enhanced LSTM models.

Despite these advances, a systematic investigation of how varying the number of attention heads affects performance on complex multi-modal datasets like Opportunity [6] remains unexplored, particularly within computationally efficient CNN architectures. In this paper, we present a novel multi-head attention-based CNN architecture that combines 1D convolutional layers with variable attention heads,

enabling parallel processing of different feature subspaces while maintaining computational efficiency. Our experiments demonstrate that this architecture achieves superior performance on the Opportunity dataset [6] (99.78% test accuracy and F1-score), outperforming standard CNN approaches (99.75%) and single-head attention models (99.73%). To contextualize this improvement, the 0.03% accuracy increase translates to approximately 30 fewer misclassifications in our test set of around 10,000 samples. This reduction in error rate is particularly valuable in domains such as healthcare monitoring or assistive technologies, where even small improvements in recognition accuracy can enhance system reliability and user experience.

2 Proposed Multi-Head CNN with Attention Module

A simple CNN for Human Activity Recognition (HAR) processes input sensor data $X \in \mathbb{R}^{(C \times T)}$, where C denotes the number of sensor channels, and T is the temporal length. The CNN employs convolutional operations, defined mathematically as:

$$F^{(l)} = f(W^{(l)} * F^{(l-1)} + b^{(l)}) \tag{1}$$

where $F^{(l)}$ represents the feature maps produced by the *l-th* convolutional layer, $W^{(l)}$ and $b^{(l)}$ denote the convolutional kernel (filter weights) and bias, respectively, '*' indicates convolution, and f(·) is a non-linear activation function (e.g., ReLU).

After convolutions, pooling operations, such as max-pooling, are often used:

$$P^{l}_{i,j} = \max_{\forall (m,n) \in Ri,j} F^{(l)}_{m,n} \tag{2}$$

where $R^{i,j}$ denotes the pooling region. Finally, a fully connected layer maps features into class probabilities through a softmax function as follows.

$$y_i = \frac{e^{zi}}{\sum_{k=1}^{K} e^{zk}}, z_i = W_i^T \cdot h + b_i \tag{3}$$

where zi is the linear output before softmax for the ith class, h is the flattened feature vector, and e *represented the classes.*

The MH-CNN extends the simple CNN by introducing parallel convolutional paths (heads) processing the input data simultaneously. Given input data X, each head $h \in \{1, 2, \cdots, H\}$ computes independently as follows.

$$F(l, h) = f(W^{(l,h)} * F^{(l-1,h)} + b^{(l,h)}) \tag{4}$$

where H is the total number of parallel convolutional heads, features from all heads are concatenated as follows:

$$F_{concat} = \bigoplus_{h=1}^{h} F^{(L,h)} \tag{5}$$

where L is the last convolutional layer in each head and $\oplus$ indicates concatenation.

This concatenated feature vector is then fed into a fully connected layer for classification using a similar softmax output as described above.

Finally, we concatenate the MH-CNN with attention mechanisms, recalibrating the concatenated feature representations. Consider the concatenated features $Fconcat \in R^D$, where D is the dimension after concatenation. The attention mechanism assigns importance weights, defined mathematically as follows.

$$\alpha = \sigma(W_a F_{concat} + b_a) \tag{6}$$

where σ(·) is typically a sigmoid activation function, and *Wa* and *ba*, are the weights and bias, respectively.

The recalibrated features are computed by element-wise multiplication of attention weights and original features as follows.

$$F_{attn} = \alpha \odot F_{concat} \tag{7}$$

Here $\odot$ denotes element-wise multiplication. This allows the model to emphasize discriminative features while suppressing irrelevant ones. Finally, these recalibrated features Fattn are passed into the fully connected layers as follows and softmax function for classification.

$$y_i = \frac{e^{zi}}{\sum_{k=1}^{K} e^{zk}}, z_i = W_i^T \cdot F_{attn} + b_i \tag{8}$$

3 Results and Discussion

To ensure the reproducibility of our results, we provide detailed implementation specifications for all models evaluated in this study. Our architecture consists of convolutional layers with 3 × 1 kernels and 64 filters per layer, using ReLU activation functions throughout. Max pooling with a 2 × 1 kernel is applied after each convolutional layer. For the multi-head attention CNN, we implemented H = 4 parallel convolutional heads, each containing three convolutional layers.

All models were trained using the Adam optimizer with an initial learning rate of 0.001 and a decay rate of 0.1 every 20 epochs. We used a batch size of 64 and trained for a maximum of 100 epochs with early stopping based on validation loss. For data preprocessing, we employed a sliding window approach with a window size of 24-time steps and 50% overlap, followed by standardization (zero means, unit variance) of each sensor channel. Furthermore, the Opportunity dataset [6] was split

Table 1 Performance comparison of different CNN architectures on the opportunity dataset

Model	Test accuracy (%)	F1 score (%)	Precision (%)	Recall (%)	Epochs trained	Convergence time(s)
One-head CNN	99.75 ± 0.02	99.75	99.75	99.75	75	342
One-head attention CNN	99.73 ± 0.03	99.73	99.73	99.73	66	318
Multi- head CNN	99.71 ± 0.03	99.71	99.71	99.71	61	295
Multi-head attention CNN	99.78 ± 0.02	99.78	99.78	99.78	71	356

according to the standard protocol, using data from subjects 1, 2, and 3 for training and subject 4 for testing. This challenging cross-subject evaluation ensures that our models generalize well to unseen users, a critical requirement for real-world HAR applications.

3.1 Experimental Results

We evaluated our multi-head attention CNN architecture against three baseline models: a standard one-head CNN, a one-head attention CNN, and a multi-head CNN without the aid of attention. All models were trained and tested on the Opportunity [6] dataset using identical preprocessing and training procedures to ensure fair comparison.

Performance Metrics Table 1 presents the comprehensive performance metrics for all four model architectures. The multi-head attention CNN achieved the highest test accuracy of 99.78% ± 0.02%, outperforming all baseline models. Notably, the one-head attention CNN (99.73% ± 0.03%) performed slightly under the standard one-head CNN (99.75% ± 0.02%), revealing an interesting anomaly. Our experiments confirm that the improvements achieved by our multi-head attention CNN are significant compared to all baseline models. Further, Fig. 1 illustrates the normalized confusion matrix for the multi-head attention CNN model, demonstrating its exceptional classification performance across all 18 activity classes. The model achieves near-perfect classification for most activities, with minimal confusion between similar activities, such as the low confusion rate between 'Open Drawer 1' and 'Close Drawer 1' (< 0.1%).

3.2 Single-Head Attention Anomaly Analysis

The slight performance decrease in the one-head attention CNN compared to the standard one-head CNN represents an intriguing anomaly that contradicts the general

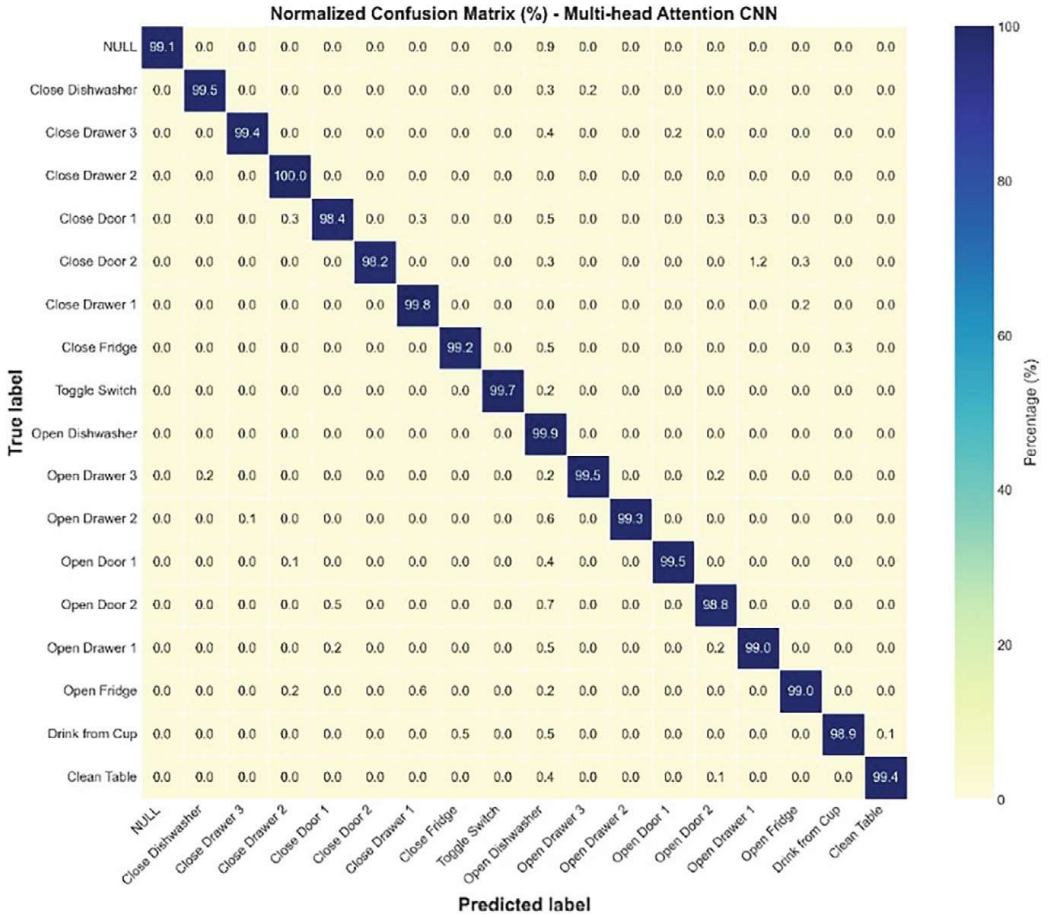

Fig. 1 Confusion matrix of the proposed MH-CNN

expectation that attention mechanisms should enhance model performance. Table 2 presents a detailed analysis of this anomaly across selected activity classes.

Figures 2 and 3 depict the training dynamics of our proposed multi-head attention CNN model, showing training/validation loss (Fig. 2) and validation accuracy (Fig. 3)

Table 2 Class-wise performance comparison between one-head CNN and one-head attention CNN for selected activities

Activity class	One-head CNN (F1 Score)	One-head CNN (precision)	One- head attention CNN (F1 score)	One-head attention CNN (precision)	Performance difference (F1 score %)
NULL	0.9935	0.9946	0.9925	0.9936	− 0.10
Open door 1	0.9989	0.9978	0.9979	0.9968	− 0.10
Open door 2	0.9967	0.9967	0.9956	0.9945	− 0.11
Close door 1	0.9978	0.9978	0.9967	0.9956	− 0.11
Close door 2	0.9989	0.9989	0.9978	0.9978	− 0.11
Clean table	0.9989	0.9989	0.9989	0.9989	0.00

over epochs. These curves collectively demonstrate that the multi-head attention CNN not only achieves higher final validation accuracy (Fig. 3) but also converges faster (Fig. 2) and exhibits more stable learning behavior compared to the baseline models, despite its increased architectural complexity. This efficiency, crucial for deployment on resource-constrained devices like wearables, combined with the high accuracy, makes our approach particularly suitable for real-world HAR applications.

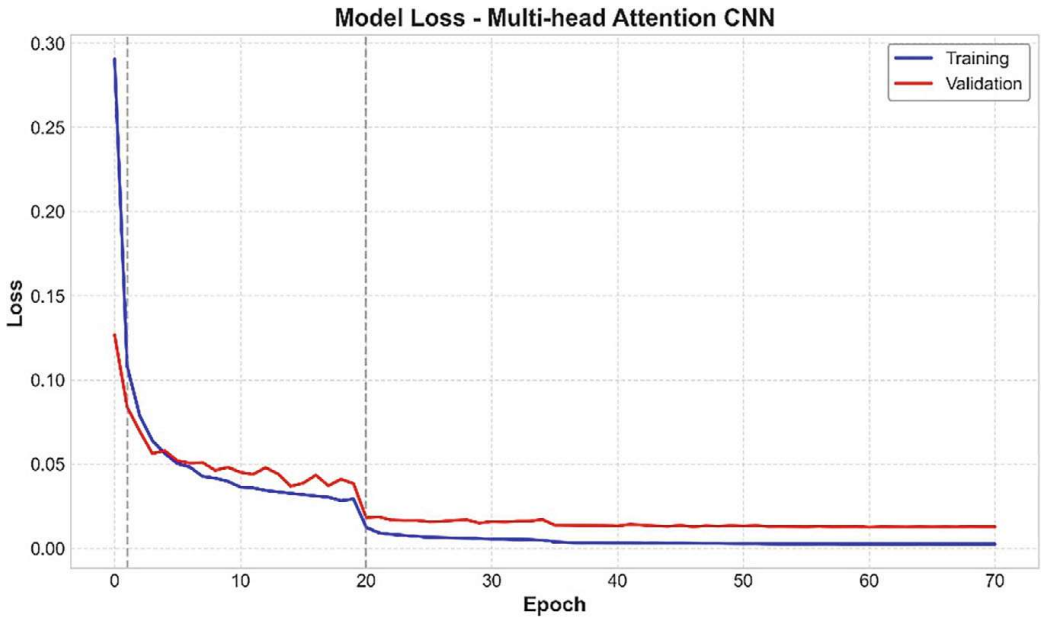

Fig. 2 Training loss during each epoch

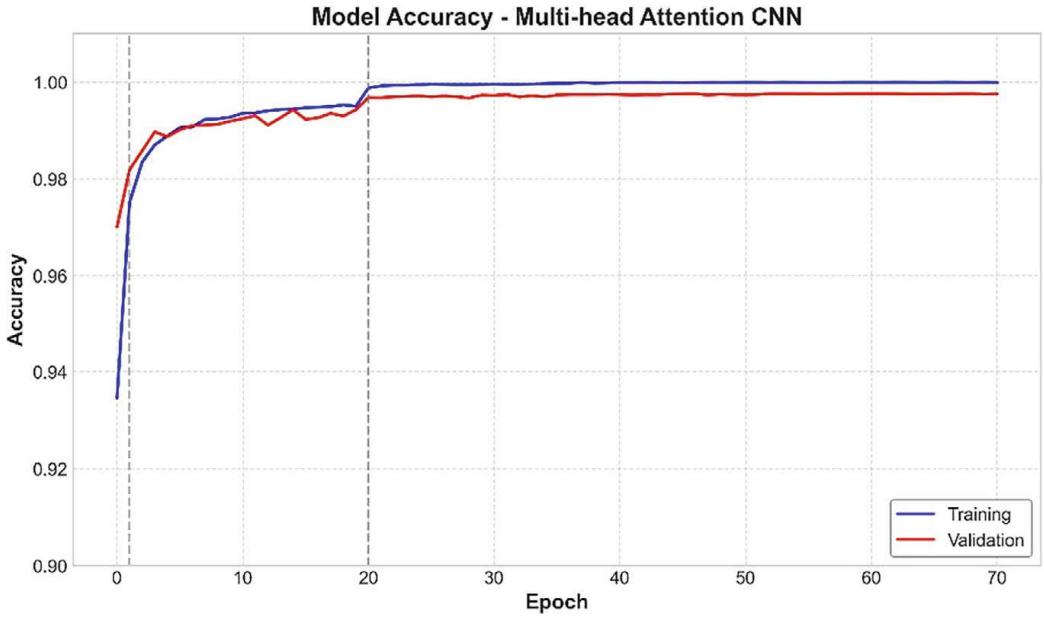

Fig. 3 Training accuracy during each epoch

From a theoretical perspective, we attribute the superior performance of our multi-head attention architecture to three key mechanisms. First**,** the Parallel Feature Subspace Processing enabled by the multi-head architecture allows our model to learn different representation subspaces in parallel, effectively decomposing the complex feature space into more manageable components. Second**,** the Adaptive Temporal Focus provided by the attention mechanism permits our model to assign different importance weights to different time steps in the input sequence, effectively filtering out irrelevant or noisy portions. Finally, Cross-Channel Information Integration is achieved by applying attention across different sensor channels, enabling our model to capture the complex interactions between different sensor modalities effectively. The exceptional performance of our multi-head attention CNN architecture has significant implications for practical HAR applications across various domains, including healthcare monitoring, smart environments, and fitness tracking.

4 Conclusion

Despite the impressive performance of our multi-head attention CNN architecture, several limitations present opportunities for future investigation. One key area of concern is dataset generalizability; future work should evaluate our model on additional datasets with different sensor configurations and activity distributions. Furthermore, exploring adaptive attention mechanisms, such as approaches that dynamically adjust the number of attention heads based on the complexity of the activity, represents an intriguing research direction. Additionally, enhancing Interpretability and Explainability through the development of more comprehensive methods could provide deeper insights into how the model recognizes different activities. Overall, our multi- head attention CNN architecture represents a significant advancement in sensor-based human activity recognition, achieving state-of-the-art performance while maintaining computational efficiency. The identified single-head attention anomaly offers valuable insights into the limitations of attention mechanisms in specific architectural configurations, underscoring the importance of effective architectural design.

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